Dynamic Systems Next Token Prediction

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1 Introduction

Large language models such as the ones behind modern chatbots (ChatGPT, Claude, Gemini, etc) work by predicting what is next based on what has come before. While many people associate this idea with text generation, nothing in it is specific to words. A variety of observations, such as sales by week, sensor readings from a machine, or prices by the minute can be cut into labelled pieces and treated in the same way. This project tests that idea on the Lorenz equations, a classic model of chaotic weather. We turn the motion of the system into a sequence of symbols, teach a next-symbol predictor with examples, and compare what it generates with the true trajectory. The results speak to questions businesses face when they forecast with sequence models like how finely to divide the data, how much history to use, and what to expect. Like weather, short-term paths

Bins per axis	V	Visited
4	64	35
16	4,096	507
64	262,244	6980

Table 1: Vocab size and number of bins visited for bins per axis $= 4, 16, 64$

are briefly predictable. However, a good model can still reproduce the overall pattern and rhythm of the system. We also check an AI assistant's advice against the known physics, since knowing when such a tool can be trusted matters as much as using it. The code in this assignment was generated by Claude Sonnet 5.5 (Anthropic).

2 The Lorenz System

The Lorenz system is given by the equations

$$\dot{x} = \sigma(y - x) \quad \dot{y} = x(r - z) - y \quad \dot{z} = xy - \beta z.$$

In this assignment, we consider the parameter choice

$$\sigma = 10 \quad r = 28 \quad \beta = \frac{8}{3}.$$

This system is chaotic. It is deterministic, in other words, the future behavior is determined by its initial conditions. However, long term prediction is difficult due to chaotic nature.

3 Tokenization

To discretize the phase space into boxes, we find the minimum and maximum of each coordinate in the data, pad with a margin, and divide the bounding box into the selection number of bins. Each box received an integer ID from its grid position

$$(i, j, k) \to i \cdot ny \cdot nz + j \cdot nz + k.$$

We take each ID to be a token. Then the vocab size, V, is given by $n_x \cdot n_y \cdot n_z$. Therefore, V increases as a cube of the number of bins per axis. However, the number of boxes actually visited does not grow as quickly because the attractor is a thin sheet inside the box. So, we consider the visited count to be the effective vocab since unvisited boxes never show up in the training data. Table 1 shows the vocab size and number of bins visited for varied bins per axis ratios.

4 Constructing Next-Token Predictor

We study 200 trajectories ($T = 100, dt = 0.05$), taking 150 for training and 50 for testing. For a predictor, we use the Markov model with $m = 1, 2, 3$ and compare the differences. We count each pair (previous m tokens, next token) in training sequences. The estimate is given by

$$\frac{\#(\text{context, b})}{\#(\text{context})}.$$

If a test context was never seen in training, the model uses the longest shorter suffix. Table 2 shows the result for varied values of m.

m	Contexts Seen	Coverage	Top-1	Top-3	Perplexity
0	1	1.0	0.018	0.053	223.7
1	500	1.0	0.541	0.892	3.19
2	1909	0.998	0.674	0.957	2.24
3	4112	0.997	0.736	0.974	1.91

Table 2: Difference in predictive power of Markov-Model predictor with varied values of m

m	Dist. Mean	¡2 units	Switches	Dwell	Fraction $x > 0$	Total Variance Dist.	Diverge t
True	0.62	0.99	58.7	1.70	0.507	0.000	NAN
1	0.62	0.99	59.6	1.69	0.490	0.040	0.25
2	0.62	0.99	57.1	1.74	0.489	0.040	0.45
3	0.62	0.99	59.3	1.68	0.499	0.036	0.62

Table 3: Comparison of generated sequences with $m = 1, 2, 3$ and true Lorenz trajectory.

4.1 Context Lengths

From our experiments on varied values of m, we see that one step of history is not sufficient, having more steps is helpful. Top-1 increases from 54.1% to 67.4% to 73.6% for $m = 1, 2, 3$ respectively. However, the gains shrink as m continues to grow. The move from $m = 1$ to 2 adds 13 points of accuracy, while the jump to $m = 3$ from $m = 2$ only adds 6. Finally, data scarcity does not present a problem for these values. $m = 3$ has 4,112 contexts, yet we still have 99.7% pf test positions having context that has been seen in testing. This is because the Lorenz system is deterministic. Therefore, only a thin set of box sequences can occur.

5 Generating New Token Trajectories

To setup, we have $V = 4096$, generate 50 sequences and 2,001 steps. Table 3 displays the results. Total Variance Dist. tells the total variance for the distance between the generated and the true token history. Diverge t is the median time until the generated path is > 5 units from the true path. Roughly 99% of the generated points lie within roughly 2 units of the true point. In other words, the generated token trajectories stay on the Lorenz attractor. In addition, for each value of m tested, the trajectories switch 56-58 times per 100 time units. The mean dwell is roughly 1.70 time units, as seen in the true trajectory. Finally, the generated trajectories have $x > 0$ $48 - 49\%$ of the time versus 50.5% in the true data. In short, each generated trajectory reproduces the two-lobe switching, as seen in Figure 1. We also note that each trajectory visits approximately 195 boxes in the last 500 steps, so none of them get stuck.

While the generated trajectories are able to reproduce the lobe-switching behavior, they do not exactly match the path of the true Lorenz system. The generated $x(t)$ separates from the true $x(t)$ almost immediately, as seen in Figure 2

6 Discretization

We use grids $4, 8, 16, 32, 64$ and study the effects on prediction and geometric information retainment. With 4 bins, we achieve a Top-1 score of 80% at $m = 3$, and the average point is 5.8 units

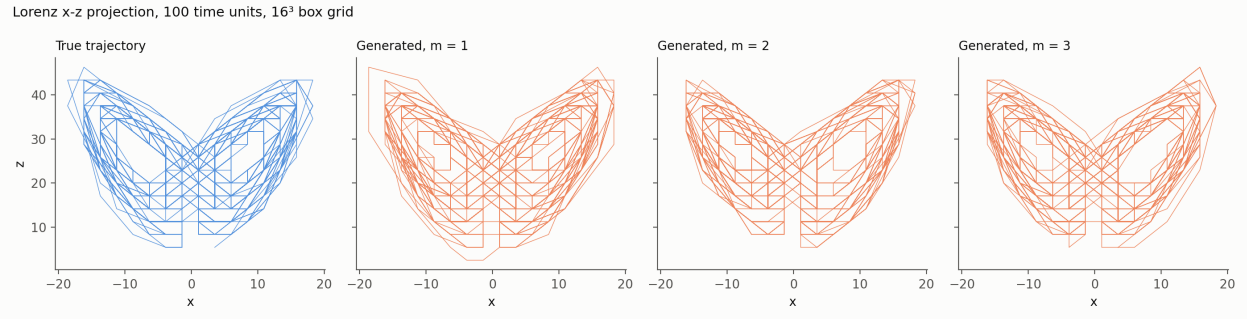


Figure 1: Projection of trajectories in the xz-plane for the true Lorenz system and the generated paths.

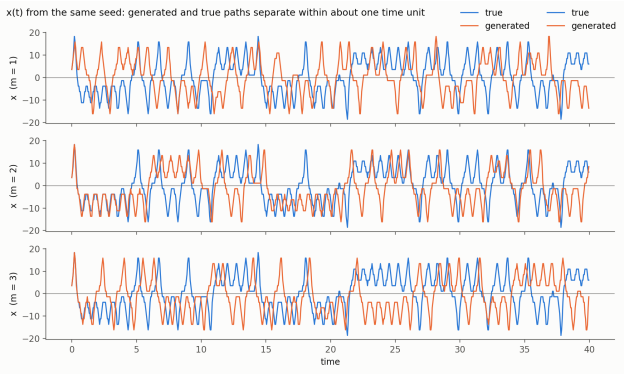


Figure 2: Plots of generated $x(t)$ versus true $x(t)$.

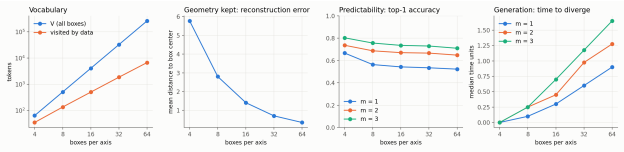


Figure 3: Results of discretization experiment for grids $4, 8, 16, 32, 64$ and $m = 1, 2, 3$.

away. This configuration switches lobes 57 times per 100 unit times, so the coarse model keeps the lobe structure but loses the finer information. In comparison, with 16 bins, Top-1 is 73.5% at $m = 3$ and the average point is 1.41 units away. Figure 3 shows plots for the size of vocabulary, geometry reconstruction error, Top-1 accuracy, and time to diverge for each grid size and value of m. We conclude that fine boxes keep the geometry and give longer faithful generation. However, accuracy decreases. Thus, a coarser grid is better for next box prediction, and a finer grid is superior for faithful trajectories.

7 LLM Consultation

The LLM referred to in this section is Claude Sonnet 5.5 (Anthropic).

7.1 Prompting and Responses

7.1.1 Prompt 1

```
For the Lorenz system (sigma=10, r=28, beta=8/3): is it dissipative,
Hamiltonian, integrable or chaotic?
Which quantities are conserved?
I generate token trajectories from a learned model;
what test should I run to see whether they drift away from the
physically admissible region?
```

7.1.2 Response

The LLM correctly identifies the Lorenz system as dissipative with divergence $-41/3 \approx -13.67$. It also identifies it is not Hamilton or integrable, it is chaotic.

7.1.3 Prompt 2

```
I discretized Lorenz into 16x16x16 boxes (one token per box, dt = 0.05)
and fit an m-step count model.
Top-1 next-token accuracy is 0.54, 0.67 and 0.74 for m = 1, 2, 3.
Generated trajectories switch lobes about 57 times per 100 time units (true: 56.8),
but separate from the true path after about 0.3 to 0.75 time units.
Why does accuracy rise with m? Is this model Markov?
What do the divergence times mean?
```

7.1.4 Response

The LLM explains the gain with m as history recovering direction and speed, also saying the box sequence is not truly Markov. It relates the 0.3-0.75 divergence to the Lyapunov time and the

along-track versus cross-track error distinction.

7.1.5 Prompt 3

```
Which of these sequences of (x, y, z) from Lorenz
(sigma=10, r=28, beta=8/3, dt = 0.05) could be a true trajectory?
Justify with the equations and numbers.
```

```
A:
(-11.0, -14.4, 25.9)
(-12.3, -13.6, 30.5)
(-12.1, -10.3, 33.6)
(-10.6,  -6.5, 33.9)
( -8.3,  -3.8, 31.9)
( -6.3,  -2.7, 29.0)

B:
(-11.0, -14.4, 25.9)
(-12.3, -13.6, 30.5)
( 13.8,  17.5, 11.2)
( 13.1,  16.9, -3.4)
( 12.0,  15.2,  9.0)
( 10.5,  13.8,  6.1)
```

7.1.6 Response

The LLM correctly identifies that A is an acceptable trajectory and B is not.

7.1.7 Prompt 4

```
I read that Lorenz has a conserved energy and is Hamiltonian,
so it shouldn't be dissipative. Are you sure?
```

7.1.8 Response

The model responds by holding its position and re-computing the divergence correctly.

7.2 Evaluation

The LLM did not confuse dissipative with Hamilton. It computed the divergence, noted that the Lyapunov exponents sum to that value, and concluded that the flow is dissipative not Hamiltonian. When I pushed back, falsely claiming that Lorenz has a conserved energy, it held its position and re-derived the divergence. Additionally, it did not name converved quantities. It listed the reflection symmetry, trapping region, equilibrium, and invariant z-axis, labeling them as symmetries and bounds rather than conservation laws. It accepted the real sequence A and rejected the fabricated sequence B with correct numbers.